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RESEARCH

Expression Profiling of Genome-Wide Identified and Characterized Trans-Isoprenyl Diphosphate Synthase Family in *Catharanthus roseus* (L.) G. Don Under Stress Conditions

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Abstract

Catharanthus roseus is known for producing anticancer monoterpenoid indole alkaloids (MIAs) vincristine and vinblastine. These alkaloids are synthesized in a complex pathway with several catalyzing steps. *Trans*-isoprenyl diphosphate synthase (TIDS) is an important family of enzymes that catalyze the initial biosynthesis steps of MIA production. However, comprehensive characterization of the TIDS gene family has not yet been done in *C. roseus*. A combination of computational and laboratory techniques was used in the current study to ascertain the role of the TIDS family under cold, hormonal, wounding, and yeast extract stress conditions. In silico analyses of the identified genes included motif and gene structure analysis, prediction of cis-acting elements, phylogenetic, evolutionary, and expression analysis. The homology modeling approach identified 4 homologs each of *geranyl diphosphate synthase* (*GPPS*) and *geranylgeranyl diphosphate synthase* (*GGPPS*), resulting in a total of 8 *TIDS* genes in the genome of *C. roseus*. Furthermore, the *Cr*TIDS proteins were clustered into four phylogenetic groups. Expression studies using qRT-PCR revealed the highest expression of *CrGPPS-01* in response to 1 g/L yeast extract (119-fold), followed by cold stress for 72 h (112-fold). The phenomenal upregulation of the *CrGPPS-01* points to the involvement of this homolog in regulating stress conditions culminating in MIA production. The results of the present study offer important insights into the *CrTIDS* gene family and a step forward in shaping the regulatory checkpoints for an improved biosynthesis of desired MIAs solely produced by *C. roseus*.

Keywords Medicinal plants · Geranylgeranyl Diphosphate Synthase (GGPPS) · Monoterpenoid Indole Alkaloids (MIAs) · Homology protein modelling · Apocynaceae

Introduction

Catharanthus roseus (L.) G. Don, a perennial species within the family Apocynaceae, is indigenous to the Indian Ocean Island of Madagascar. Widely cultivated as an ornamental plant across various regions, this species is highly valued for its medicinal properties. It has been traditionally and pharmacologically utilized to treat numerous conditions, including diabetes, hypertension, and cancer (Pham et al. 2020). It is the sole source of vincristine and vinblastine—two important anti-cancerous compounds, in addition to several bioactive MIAs like serpentine and ajmalicine that are used to cure circulatory disorders. *C. roseus* is reportedly known to produce 4000 alkaloids and over 130 terpenoid indole alkaloids (Zhu et al. 2014). The biosynthesis of MIAs comprises 50 known biosynthetic events consisting of enzymes, regulators, genes, and inter- or intracellular transporters (Zhao et al. 2013).

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The isoprenyl diphosphate synthase (IDS) gene family encompasses both cis- and trans-isoprenyl diphosphate synthases (TIDS). The TIDS are responsible for generating short-chain isoprenyl diphosphates (ranging from C10 to C50), whereas cis-isoprenyl diphosphate synthases are typically associated with the production of long-chain isoprenyl diphosphates exceeding C50 (Jia and Chen 2016). Key members of the TIDS group include GPPS, farnesyl diphosphate synthase (FPPS), and GGPPS (Beck et al. 2013). These enzymes play a crucial role in catalyzing the formation of various plant terpenoids and terpenoid-derived alkaloids such as MIAs. The terpene biosynthesis involves a condensation reaction between IPP (isopentyl diphosphate) and DMAPP (dimethylallyl diphosphate). The synthesis of IPP and DMAPP is facilitated by two distinct metabolic pathways: the mevalonic acid (MVA) pathway, which predominantly operates in the cytoplasm of eukaryotic cells, and the methylerythritol phosphate (MEP) pathway, which is active in plastids and is common in plants and some bacterial species. In these pathways, IPP and DMAPP undergo sequential condensation reactions, mediated by various TIDS enzymes, resulting in the production of increasingly complex isoprenoid molecules. GPPS catalyzes the formation of geranyl diphosphate (GPP), a 10-carbon monoterpene precursor, through the condensation of IPP and DMAPP. FPPS facilitates the synthesis of farnesyl diphosphate (FPP), a 15-carbon isoprenoid, from IPP and DMAPP. FPP contains a 15-carbon isoprenoid chain and is a precursor for sesquiterpenes, while GGPPS synthesizes 20-C geranylgeranyl diphosphate (GGPP) from IPP and DMAPP. GGPP serves as a precursor for diterpenes, polyterpenes, and photosynthesis-related isoprenoids like carotenoids and chlorophylls (Kirby and Keasling 2009).

Members of the IDS gene family have been studied in numerous organisms, including fungi such as yeast (Jiang et al. 1995) and *Neurospora* (Galagan et al. 2003), as well as bacteria (Feng et al. 2020), mammals (Kainou et al. 1999), plants (Okada et al. 2000), and insects (Hojo et al. 2007). Among plant species, *Arabidopsis thaliana* contains 17 *TIDS* genes, including 12 *GGPPS*, 2 *FPPS*, and 1 gene encoding a small subunit of *GPPS* (Coman et al. 2014), while the nuclear genome of *Chimonanthus praecox* L. contains 2 heterodimeric *GPPS.SSU*, 1 homomeric *GPPS*, and 2 *GGPPS* (Kamran et al. 2020). Other species in which *TIDS* gene family has been characterized include cotton (Ali et al. 2020; Zhang et al. 2022; Wen et al. 2023), *Prunus persica* (Li et al. 2022), *Liriodendron chinense* (Cao et al. 2023), *Daucus carrota* (Keilwagen et al. 2017), *Ocimum sanctum* (Kumar et al. 2018), *Psidium cattleyanum* (Canal et al. 2023), *Dendrobium officinale* (Yu et al. 2020), *Cinnamomum camphora* (Yang et al. 2021), *Mentha longifolia* (Chen et al. 2021), *Brassica oleracea* L. (Yan et al. 2023), *Salvia miltiorrhiza* (Ma et al. 2012), and *Nelumbo nucifera* (Dong et al. 2023a, b). These studies reflect that most *TIDS* genes encode homomeric synthases; however, GPPS protein exists in both forms (Nagegowda 2010). The homomeric GPPS proteins have also been identified from various angiosperms (Bouvier et al. 2000; van Schie et al. 2007) as well as gymnosperms (Schmidt et al. 2010).

Both GPPS and GGPPS proteins are integral to diverse biosynthetic pathways, with their functional diversity being closely associated with subcellular localization and spatiotemporal expression across plant species. For example, plastid-localized *At*GGPPS7 and *At*GGPPS11 are involved in the biosynthesis of photosynthesis-related isoprenoids, while *At*GGPPS1, located in mitochondria, contributes to gibberellin production. Similarly, *At*GGPPS3 and *At*GGPPS4 are restricted to the endoplasmic reticulum. Notably, the *At*GGPPS genes are expressed ubiquitously across plant organs, with pronounced expression in roots and siliques (Beck et al. 2013). Beyond plants, GGPPS proteins have significant implications for human health, including their roles in disease treatment and potential applications in cancer therapies (Muehlebach and Holstein 2023). GPPS proteins primarily synthesize monoterpene precursors. Interestingly, the expression of *Antirrhinum majus GPPS.SSU* alone in *E. coli* does not yield detectable prenyltransferase activity; however, co-expression with *GPPS.LSU* forms an active heterodimer capable of synthesizing GPP. Studies suggest that the small subunit is critical for regulating monoterpene biosynthesis (Tholl et al. 2004). In tobacco, expression of *A. majus GPPS.SSU* converts endogenous inactive GPPS protein into an active form, enhancing its catalytic activity and monoterpene production (Orlova et al. 2009). In tomato, co-expression of *A. majus GPPS.SSU* gene with the *geraniol synthase* gene from *Ocimum basilicum* leads to a three-fold increase in monoterpene production (Gutensohn et al. 2013). In *C. roseus*, three genes have been identified that encode GPPS.SSU, GPPS.LSU, and homomeric GPPS proteins, all of which are implicated in monoterpenoid biosynthesis (Rai et al. 2013).

Recent studies have identified *wall-associated kinases* (Ahmed et al. 2024), *BES1* (Dong et al. 2023a, b), *NAC* TF (Ahmed et al. 2023), and *TCP* (Hao et al. 2023) gene families in *C. roseus*. All these studies contributed to better insights into the regulation of MIAs in *C. roseus*. Based on this, it was hypothesized that the identification and characterization of the TIDS enzyme family in *C. roseus* may contribute towards addressing the low concentration problem of MIAs in *C. roseus*. Although the TIDS gene family has been extensively studied in several plant species, its identification and analysis in *C. roseus* have not been reported yet. Therefore, in this study, a comprehensive genome-wide identification and analysis of the TIDS gene family in *C. roseus* was performed.

Materials and Methods

Sequence Retrieval

Medicinal plant genomics resource (MGPR) was used to retrieve the *C. roseus* genome, proteome, and CDS sequences (Supplementary Material). *Arabidopsis thaliana* GPPS protein (NP_001031483.1), a member of the TIDS family, was used to BLASTp against the *C. roseus* proteome. Additionally, the HMM profile of the TIDS domain (prenyl synthase domain PF00348) was obtained from the Pfam database (Finn et al. 2013) and used to identify TIDS members via HMMER3.0 (e-value, $1e^{-2}$). The obtained sequences of the potential TIDS members were submitted to SMART (Schultz et al. 1998), Pfam, and NCBI-CDD batch search (Lu et al. 2020) to validate the presence of prenyl synthase domains. A total of 8 TIDS genes were selected and submitted to ExPASy-ProtParam (Gasteiger et al. 2005) to calculate the physicochemical properties. WoLF PSORT was used to predict sub-cellular localization (Horton et al. 2007).

Gene Structure, Motif, and Promoter Analysis

The intron–exon organization of *CrTIDS* genes was predicted through a gene structure display server (Hu et al. 2015), while MEME software (Bailey et al. 2015), set at a maximum of 10, was used for motif analysis with default parameters. The identified motif patterns were visualized using TBtools software (Chen et al. 2020). The 1500 bp promoter region of the *CrTIDS* genes was obtained from the MPGR. These sequences were submitted to the PlantCare database (Lescot et al. 2002) to identify cis-acting elements, and the results were displayed using TBtools software.

Evolutionary Analysis of *CrTIDS* Proteins

The TIDS protein sequences of *Arabidopsis thaliana*, *Gossypium hirsutum*, *Solanum lycopersicum*, and *Brassica oleracea* were retrieved from Phytozome and aligned with *Cr*TIDS using the ClustalW program. The phylogenetic tree was constructed by the maximum likelihood method using MEGA7 with 1000 bootstrap replicates. (Kumar et al. 2018).

Chromosomal Localization and Duplication Analysis of *CrTIDS* Genes

Chromosomal positions and sizes for *CrTIDS* genes were obtained using the *C. roseus* genome browser available on NCBI. The distribution of *CrTIDS* genes across the chromosomes was visualized using TBtools. Gene duplication events were identified with the MCScanX tool (Wang et al. 2012), and the relationships between paralogous TIDS genes were illustrated using a dual synteny plot in TBtools. The Ka/Ks ratio for duplicated gene pairs was calculated with the Ka/Ks calculator in TBtools to assess selective pressure (Chen et al. 2020).

Gene Expression Analysis

The expression profiles of *TIDS* genes in *C. roseus* were evaluated using expression data retrieved from MPGR and CathaCyc. Expression data for *CrTIDS* genes were gathered across four different organs: roots (SRR122254), stems (SRR122253), flowers (SRR122239), immature leaves (SRR122252), and mature leaves (SRR122251). Additionally, expression patterns were examined in *C. roseus* cell suspension cultures (SRR122240-SRR122250) and hairy root cultures (SRR122256-SRR122261) treated with various elicitors. The FPKM values obtained were normalized to log2 values, and heat maps were generated using TBtools (Chen et al. 2020). The Cytoscape V 3.8.2 software was employed to create a co-expression network between biosynthetic genes and CrTIDS genes.

Plant Material, Treatment, and Experimental Design

Catharanthus roseus seeds were acquired from Pan America Seeds Company (USA). The seeds were surface sterilized with 70% ethanol for 30 s followed by 5% NaOCl for 1 min to remove any contamination, and then cultivated in pots (10 cm diameter × 5 cm height) containing a potting mixture consisting of peat moss, coco peat, and vermicompost in a 1:1:1 ratio. The pots were covered with aluminum foil for a few days to provide the necessary dark period for germination and maintained in the growth chamber at 26 °C with 16:8 light/dark cycles. In a completely randomized design, two-month-old seedlings were selected for stress treatments, including cold stress, wounding, drought stress, and yeast extract treatment, while hormonal stress included salicylic acid treatment. The plants were incubated at 4℃ for cold stress for 24, 48, and 72 h. Mechanical wounding of the third leaf was performed carefully with the help of forceps. The yeast extract solutions (1 g/L and 2 g/L) and salicylic acid solutions (1 mM and 3 mM) were sprayed on foliar parts. Samples were collected after 24, 48, and 72 h of treatment. Fresh samples were preserved immediately in liquid nitrogen.

RNA Extraction, cDNA Synthesis, and qRT-PCR

Total RNA was extracted from the control and treated plant samples using TRIzol reagent (Invitrogen). The quantity and purity of extracted RNA were checked using a NanoDrop (Thermo Scientific Nanodrop 1000 spectrophotometer), while its quality was checked on a 1.5% agarose gel. Complementary DNA was prepared from extracted RNA of *C. roseus* seedlings by using ABScript III RT Master Mix for qPCR with genomic DNA remover (ABclonal Technology). The newly synthesized cDNA was subjected to real-time PCR for expression analysis. Real-time PCR was performed in CFX 384 Touch real-time PCR (Bio-Rad Laboratories Inc.) using 2× Universal SYBR Green PCR master mix. Targeted genes were amplified using gene-specific primers, while *RPS9* was used as an internal control.

Statistical Analysis

The data were presented as mean ± SD. One-way ANOVA followed by Tukey HSD was used to indicate significant differences in the mean values.

Results

Identification of TIDS Homologs in *C. roseus*

The HMM and BLASTp methods, used to identify TIDS proteins, revealed a total of 4 GPPS and 4 GGPPS together making a total of 8 TIDS in *C. roseus*. The obtained proteins were confirmed to contain the iso-prenyl domain using various databases and were renamed from *Cr*GPPS/*Cr*GGPPS-01 to *Cr*GPPS/*Cr*GGPPS-08. The physiochemical properties of *Cr*TIDS homologs are shown in Table 1. The *Cr*TIDS protein ranges from 261 amino acids (*Cr*GPPS-01) to 889 amino acids (*Cr*GPPS-02) with an average length of 525 amino acids. These proteins have molecular weights from 28.067 kDa (*Cr*GPPS-01) to 97.276 kDa (*Cr*GPPS-02). All *Cr*TIDS were acidic (pI < 7), averaging a pI of 5.77. The hydrophilic indices of *Cr*TIDS were all less than 0, suggesting they are hydrophilic. The subcellular localization predicted by WoLF PSORT revealed that 4 *Cr*TIDS were localized in the chloroplast, while 1 in each nucleus, cytoplasm, endoplasmic reticulum, and plasma membrane.

Phylogenetic Analysis and Classification *C. roseus* TIDS

To investigate the phylogenetic relationships of *Cr*TIDS proteins, a maximum likelihood phylogenetic tree was constructed using the full-length protein sequences of 8 TIDS proteins from *C. roseus*, along with 15 from *Arabidopsis thaliana*, 6 from *Gossypium hirsutum*, 6 from *Solanum lycopersicum*, and 7 from *Brassica oleracea*. The TIDS proteins were categorized into four groups based on clustering results (Fig. 1). Among these, group IV emerged as the largest cluster, comprising 16 TIDS proteins, including 3 *Cr*TIDS (*Cr*GPPS-05, *Cr*GGPPS-06, and *Cr*GGPPS-08), alongside 5 proteins from *G. hirsutum*, 4 from *A. thaliana*, 2 from *B. oleracea*, and 2 from *S. lycopersicum*. Group I contained 13 proteins, with a single *Cr*TIDS member (*Cr*GPPS-04). Group II, the smallest cluster, included 6 proteins, of which 4 belonged to the *Cr*TIDS family, with the remaining 2 from *S. lycopersicum*. In contrast, group III consisted exclusively of 7 TIDS proteins from *A. thaliana* and one from *B. oleracea*, with no *Cr*TIDS members present in this group.

Intron–Exon Organization and Conserved Motif Analysis

To understand the structural evolution of the *TIDS* gene family in the *C. roseus* genome, the intron–exon organization of the genes was analyzed using the Gene Structure Display Server (Fig. 2). The *TIDS* genes exhibited considerable variation in length, ranging from 785 bp to 11,608 bp. The exon count also varied significantly among the genes, ranging from 1 to 15 exons. Interestingly, all three members of group III were found to contain a single exon, suggesting a simplified gene structure. In contrast, group II members typically had three exons separated by two introns. Members of group I, such as *CrGPPS-04* and *CrGPPS-05*, exhibited a more

Table 1 Summary of identified *Cr*TIDS members from *C. roseus*

Homolog ID	Name	Protein length	Molecular weight	pI	GRAVY	Aliphatic index	Sub-cellular localization
CRO_T031389	*Cr*GPPS-01	261	28,067.39	5.22	0.116	103.18	Cytoplasm
CRO_T027399	*Cr*GPPS-02	889	97,276.5	6.34	− 0.058	95.86	Chloroplast
CRO_T025194	*Cr*GGPPS-03	801	88,000.44	5.2	− 0.083	95.89	Nucleus
CRO_T025571	*Cr*GPPS-04	492	54,239.35	5.86	0.095	101.89	Plasma membrane
CRO_T031389	*Cr*GPPS-05	693	76,011.65	5.79	0.041	102.15	Endoplasmic reticulum
CRO_T006133	*Cr*GGPPS-06	371	40,202.35	5.51	0.028	99.43	Chloroplast
CRO_T008089	*Cr*GGPPS-07	337	36,500.61	5.63	− 0.123	83.12	Chloroplast
CRO_T012211	*Cr*GGPPS-08	362	39,463.82	6.67	− 0.003	93.65	Chloroplast

GRAVY grand average value of hydrophobicity

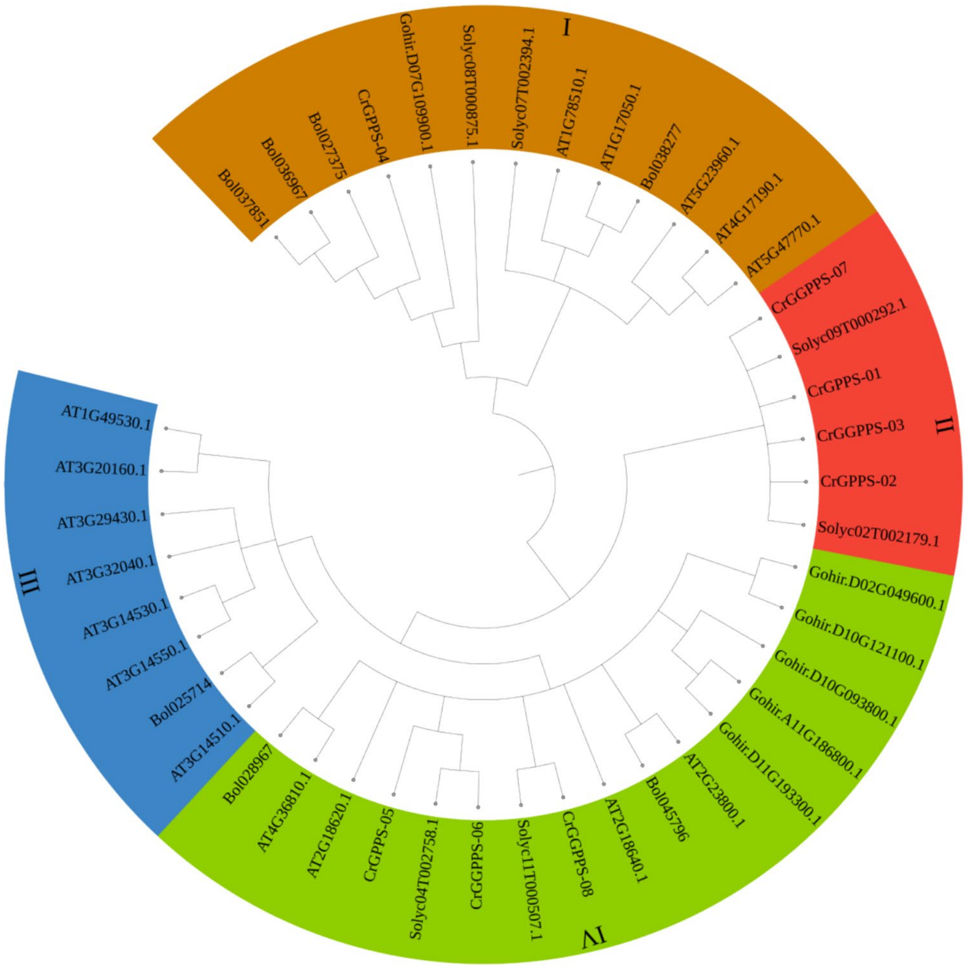

Fig. 1 Phylogenetic analysis of TIDS proteins from *C. roseus*, *A. thaliana*, *G. hirsutum* (Gohir), *S. lycopersicum* (Solyc), and *B. oleracea* (Bol). A maximum likelihood phylogenetic tree of 43 TIDS proteins was constructed using MEGA 7.0

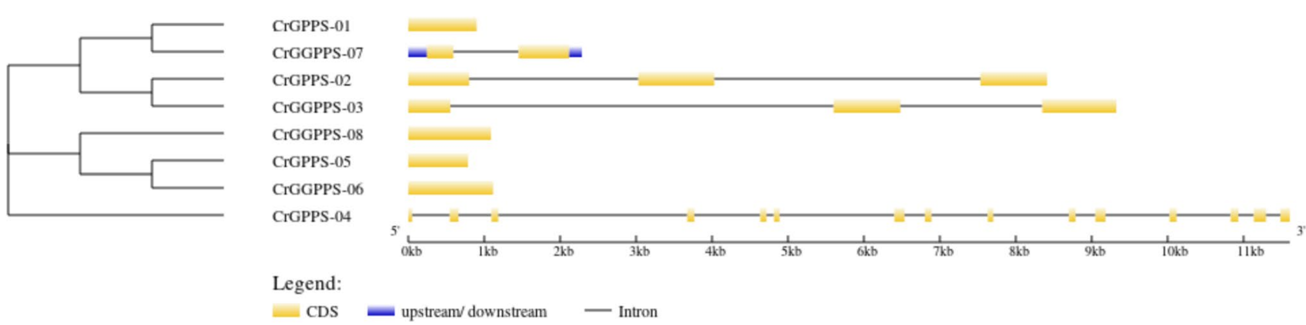

Fig. 2 The gene structure of *Cr*TIDS genes. The black line indicates introns while exons are represented with yellow boxes. The UTRs are represented by blue boxes

complex gene structure, with 15 and 13 exons, respectively. These findings indicate distinct structural differences among the groups, potentially reflecting functional specialization or evolutionary divergence within the TIDS gene family.

A total of 10 distinct conserved motifs were present in *Cr*TIDS (Fig. 3). The *Cr*TIDS proteins exhibited a notable degree of similarity in their motif composition, with motifs 2, 6, 7, and 8 being prevalent across the majority of these genes, suggesting a conserved evolutionary trait. Members of group I were found to possess between 4 to 5 motifs, indicating a diverse yet interconnected structural framework. In group III, except for *CrGPPS-01*, which lacked motif 3, the remaining two members, *CrGGPPS-08* and *CrGGPPS-06*, contained 8 similar motifs. Additionally, motifs 9 and 10 were uniquely identified in two specific members of group II, *CrGPPS-02* and *CrGGPPS-03*. The presence of these

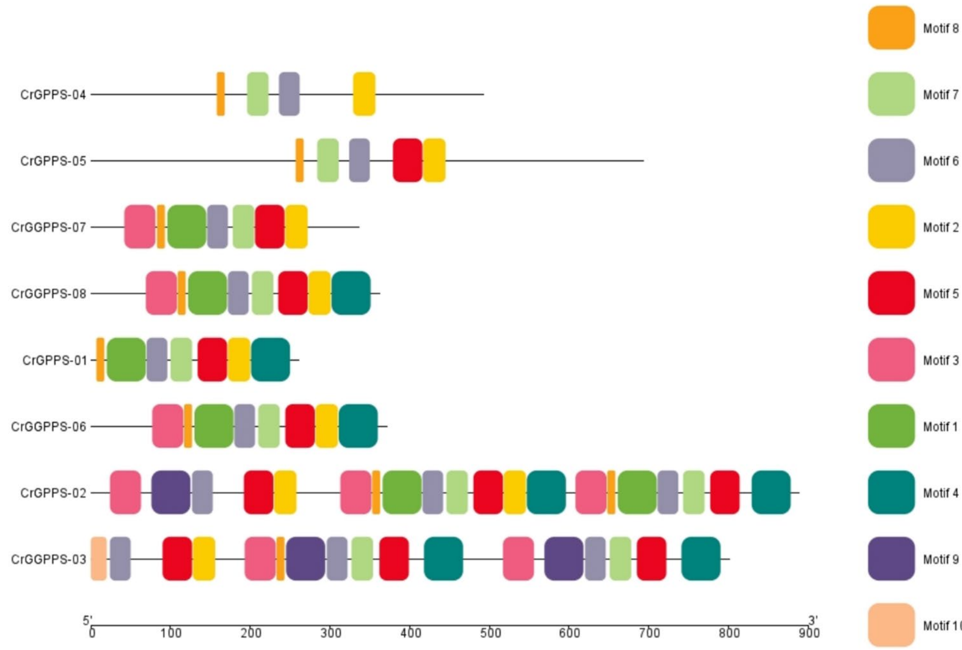

Fig. 3 Identified conserved motifs in *CrTIDS* genes. Each motif is characterized by a distinct color

exclusive motifs suggests unique functionalities or regulatory mechanisms that could be pivotal for further research.

Chromosomal Location, Gene Duplication, and Synteny Analysis

The analysis of chromosomal locations offers significant insights into the distribution of the eight *TIDS* genes in *C. roseus*. The findings indicated that the *CrTIDS* genes were predominantly located on chromosome 1, which harbors three genes, highlighting a notable concentration in this region. Chromosome 8 contained two *CrTIDS* genes. Additionally, chromosomes 3, 5, and 6 each featured one *CrTIDS* gene, while chromosomes 2, 4, and 7 do not possess any *CrTIDS* genes (Fig. 4).

Syntenic blocks within the *C. roseus* genome were systematically analyzed to enhance our understanding of the relationships among the *CrTIDS* genes and to uncover potential gene duplication events (Fig. 5). In total, three *CrTIDS* pairs were identified, resulting in six duplicated genes. Notably, one pair, *CrGPPS-02*/*CrGGPPS-03*, appeared to be tandemly duplicated, while the other two pairs, *CrGPPS-01/CrGGPPS-07* and *CrGPPS-05/CrGPPS-06*, had undergone segmental duplication. To explore the evolutionary

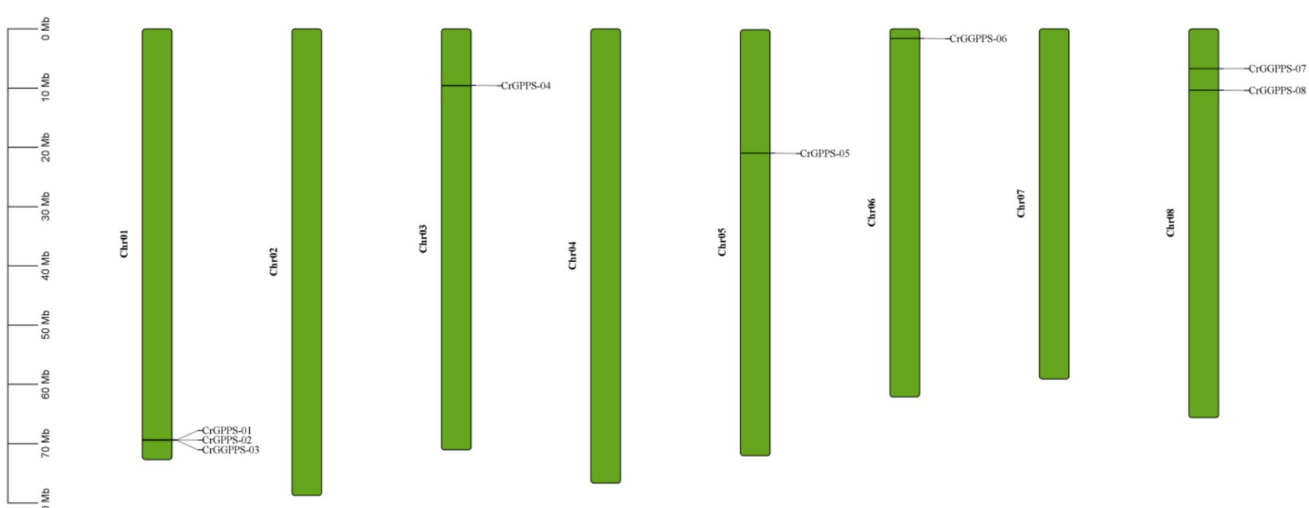

Fig. 4 Distribution of *CrTIDS* genes on *C. roseus* chromosomes

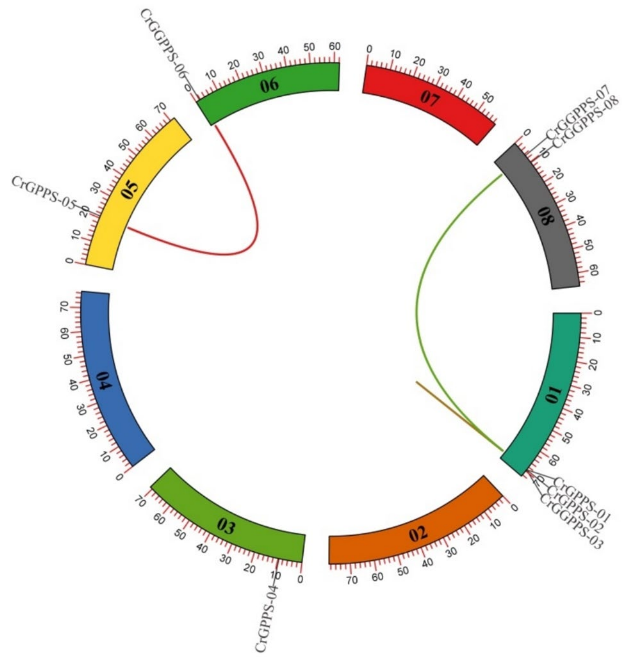

Fig. 5 Circos graph showing genes' chromosomal location and duplicated gene pairs. Colored lines indicate the duplication events

pressures shaping the *Cr*TIDS gene family, the Ka/Ks ratios for these gene pairs were calculated (Table 2). A Ka/Ks ratio between 0 and 1 suggests the presence of purifying selection. Remarkably, all gene pairs demonstrated Ka/Ks values below 1, with the *CrGPPS-05/CrGPPS-06* pair exhibiting the highest value of 0.39518437.

Promoter Analysis of *CrTIDS* Genes

In this study, a 1500 bp promoter of *CrTIDS* genes was analyzed using the PlantCARE database to understand their transcriptional regulation as well as their prospective functions. The p.romoter regions of eight *CrTIDS* genes were analyzed, revealing a significant number of cis-acting elements. Core promoter elements, such as the TATA-box and CAAT-box, essential for RNA polymerase binding and transcription initiation, were identified in all *CrTIDS* genes. Stress-responsive elements included motifs associated with low-temperature responses, hypoxia (GC-motif and ARE), and drought (MBS). Defense-related cis-elements, such as wound-inducible motifs (WUN motifs), W-box, WRE3, and TC-rich repeats, were also predicted by the tool in the promoter of selected genes. MYB binding sites were present in the promoter regions of all *CrTIDS* genes. Similarly, the MYC binding sites were commonly observed in the promoter regions of the selected genes. Light-responsive elements, including G-Box (CACGTC), Sp1 (GGGCGG), and Box-4 (ATTAAT), were identified, along with developmental motifs such as the AAGAA-motif (endosperm-specific expression) and CAT-box (meristem expression-related). The promoter regions also contained several hormone-responsive elements, including those for abscisic acid (ABRE), auxin (TGA-element), ethylene, salicylic acid (TCA-elements), gibberellin (GARE-motif and P-box), and methyl jasmonate (TGACG-motif and CGTCA-motif). Surprisingly, the promoter sequence *CrTIDS-08* gene also contained cadmium-responsive AP-1 CARE (Fig. 6).

Expression Analysis and Co-expression Network Construction

To investigate the transcription patterns of *CrTIDS* genes, a comprehensive analysis of tissue-specific expression profiles was conducted across various organs, including flowers, roots, stems, immature leaves, mature leaves, and sterile seedlings. The resulting expression data facilitated the generation of heat maps utilizing TBtools software. The findings revealed that the expression of *CrTIDS* genes was prevalent in most tissue types. This observation suggests a significant role for these genes in the regulation of plant growth and development (Fig. 7a). The *CrGGPPS-08* showed no expression in any tissue. Additionally, certain *TIDS* genes demonstrated tissue-specific expression, indicating their potential involvement in particular physiological processes. For example, transcripts of *CrGGPPS-03* were exclusively found in stem and sterile seedling samples, while the expression of *CrGPPS-02* was notably higher in flower, root, and leaf samples compared to that in stem and seedlings. Among the eight *CrTIDS* genes analyzed, *CrGGPPS-06* exhibited the highest expression levels across all organs, except for flowers, where transcripts of *CrGPPS-05* were higher.

In addition to various tissues, the transcript level of *CrTIDS* genes was also evaluated in cell suspension and hairy root cultures treated with methyl jasmonate (MJ) and yeast extract (Fig. 7b). Out of 8 genes, only *CrGGPPS-08* genes showed no expression (FPKM values were all < 1) in any sample. *CrGPPS-02* showed expression only in wild-type hairy root samples, while the expression of *CrGPPS-03*

Table 2 Ka/Ks values of *CrTIDS* duplicated gene pairs. Ka (non-synonymous substitution rate) and Ks (synonymous substitution rate)

Duplicated genes	Ka	Ks	Ka/Ks	Mode of duplication
CrGPPS-05/CrGPPS-06	0.75231029	1.222296944	0.615489	Segmental
CrGPPS-01/CrGGPPS-07	0.127325946	2.339058792	0.054435	Segmental
CrGPPS-02/CrGGPPS-03	0.239332087	0.732238576	0.32685	Tandem

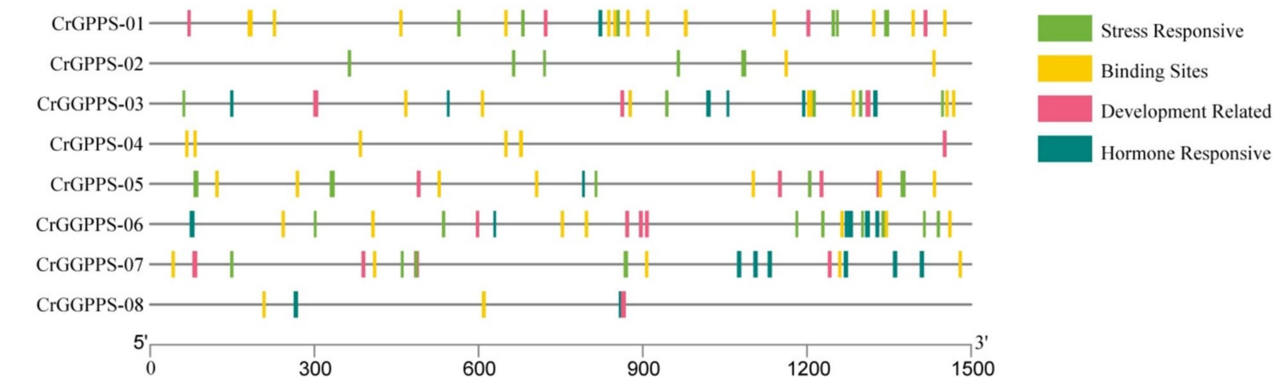

Fig. 6 Cis-acting elements present in the promoter of *CrTIDS* genes

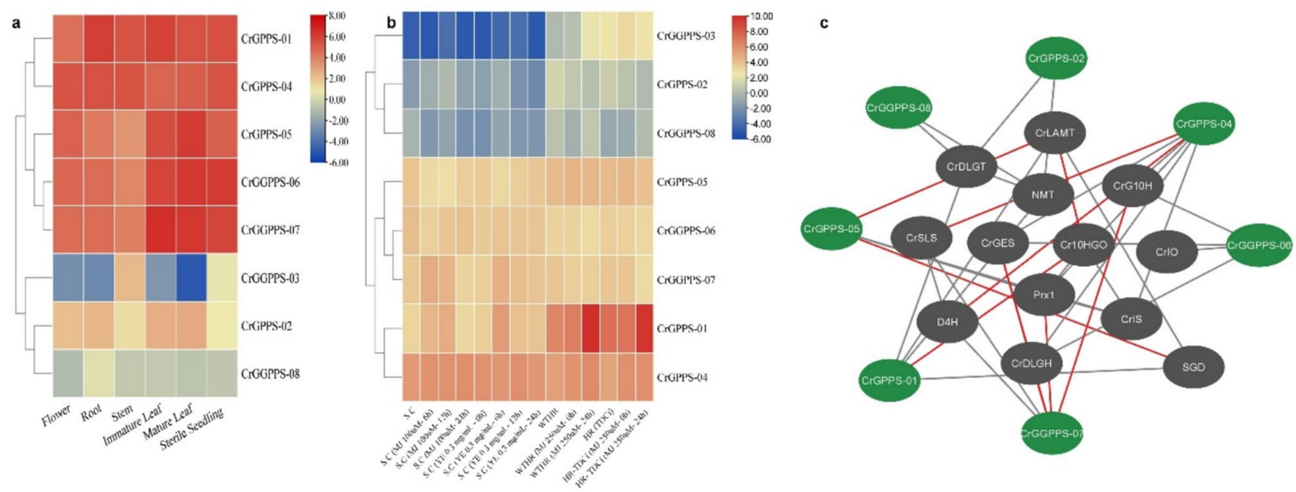

Fig. 7 **a** Heat map showing the expression pattern of *CrTIDS* genes in flower, roots, stem, immature leaf, mature leaf, and sterile seedling. **b** Expression analysis of *CrTIDS* genes in different cultures presented as a heatmap. S.C.—suspension culture, WTHR—wild-type hairy root, HR—hairy root, MJ—methyl jasmonate, and YE—yeast extract. **c** A co-expression network was constructed using Cytoscape. Nodes represent genes by nodes while edges represent interactions (positive interactions—grey edges, negative interactions—red edges)

was significant in methyl jasmonate-treated hairy root samples and hairy roots transformed with TDCi. The remaining five genes showed expression in all samples. In suspension culture media, methyl jasmonate elicitation (100 μM) for 12 h was the most effective treatment for increasing the transcription of *CrTIDS* genes, compared to 6 h and 24-h treatments. However, the transcription of *CrGPPS-05* and *CrGGPPS-06* was downregulated in response to methyl jasmonate treatment, as compared to control suspension culture media. In comparison, yeast extract elicitation was more effective in inducing the expression of all genes, especially 6 h of exposure time produced better results than 12 h and 24 h of exposure.

In wild-type hairy root cultures, methyl jasmonate treatment amplified expression of five *CrTIDS* genes, while three genes were downregulated. Moreover, elicitation with 250 μm methyl jasmonate for 24 h was most effective, particularly for *CrGGPPS-03* and *CrGPPS-01* genes that showed a significant increase in response to this treatment. Methyl jasmonate elicitation also increased expression of seven *CrTIDS* in hairy roots with inactive tryptophan decarboxylase (TDCi).

A gene co-expression network was constructed between 8 *CrTIDS* genes and 13 monoterpenoid indole alkaloids biosynthetic genes. Results showed that only *CrGGPPS-03* showed no interaction with any MIA biosynthetic pathway genes, while the remaining genes showed interaction. The *CrGPPS-02*, *CrGGPPS-06*, and *CrGGPPS-08* showed only positive interactions with the targeted genes, while the remaining genes showed both positive as well as negative correlations. *CrGPPS-04* showed a positive correlation with 5 MIA genes, while *CrGGPPS-06* showed a positive correlation with 4 genes. Moreover, *CrIO* (iridoid oxidase) only positively correlated with *CrGGPPS-06* (Fig. 7c).

Validation of Expression Pattern of *CrTIDS* Genes Under Stress Conditions

The analysis of available gene expression showed that *CrGPPS-01* (*CrGPPS LSU*) and *CrGPPS-04* (homomeric *GPPS*) play an important role in the regulation of biosynthesis of monoterpene indole alkaloids. To validate this, the expression pattern of *CrGPPS-01* and *CrGPPS-04* genes was analyzed in response to various stress and hormonal treatments using qRT-PCR. Results showed that the expression of *CrGPPS-01* was significantly upregulated in response to cold stress (a 112-fold increase compared to control plants). Similarly, foliar application of yeast extract and wounding led to 119-fold and 27-fold increases in *CrGPPS-01* expression, respectively. However, the *CrGPPS-01* gene showed only a sixfold increase in response to high concentrations of SA, particularly the 3 mM treatment. In comparison, the *CrGPPS-04* gene showed only a twofold induction in response to cold stress, while a threefold increase in response to wounding. The maximum induction of the *CrGPPS-04* gene was observed in samples treated with 2 g/L yeast extract (ninefold induction) and 3 mM salicylic acid (sevenfold induction) (Fig. 8).

Discussion

Catharanthus roseus, commonly known as the Madagascar periwinkle, is a remarkable medicinal herb that has gained significant attention due to its exceptional ability to produce valuable pharmaceutical bioactive compounds, particularly the terpenoid indole alkaloids (Lahare et al. 2020). This plant is especially renowned for its potent anticancer properties, which have spurred extensive scientific research and studies aimed at understanding its beneficial effects. The biosynthesis of terpenoid indole alkaloids (TIAs) within *C. roseus* is a complex and intricate pathway. This pathway is intricately regulated by a diverse array of transcription factors, enzymes, and metabolic intermediates that work synergistically to ensure optimal production of these medicinal compounds (Zhao et al. 2013). A crucial component of this biosynthetic process is TIDS, which belongs to a vital class of enzymes. Among this gene family are the *GPPS* and *GGPPS*. Each of these enzymes play a distinctive role in catalyzing the synthesis of specific terpene precursors. GPPS enzyme is especially noteworthy for its function in generating monoterpene precursors, which are fundamental building blocks in various natural products. On the other hand, the enzyme GGPPS is responsible for producing diterpene precursors and is also involved in the synthesis of essential primary metabolites such as chlorophylls, carotenoids, and gibberellins, all of which play critical roles in plant growth and development (Barja et al. 2021). *GGPPS* is more common in plants as compared to *GPPS*, with *GPPS* believed to have originated from *GGPPS* (Song et al. 2023). Most of the TIDS enzymes exist as homodimers; however, GPPS exists in both forms. The heteromeric GPPSs are frequently found in species that produce adequate number of monoterpenes such as *Mentha piperita*, *Antirrhinum majus*, *Clarkia breweri*, and *Humulus lupulus* (Burke et al. 1999; Tholl et al. 2004; Wang and Dixon 2009). The heteromeric GPPS protein consists of two distinct subunits: a non-catalytic small subunit (SSU) and a large subunit (LSU). The LSU may either be catalytically inactive (Burke et al. 1999) or capable of functioning independently as a GGPPS (Tholl et al. 2004; Wang and Dixon 2009). The interaction between the SSU and LSU is crucial, as it forms an active heteromeric GPPS complex.

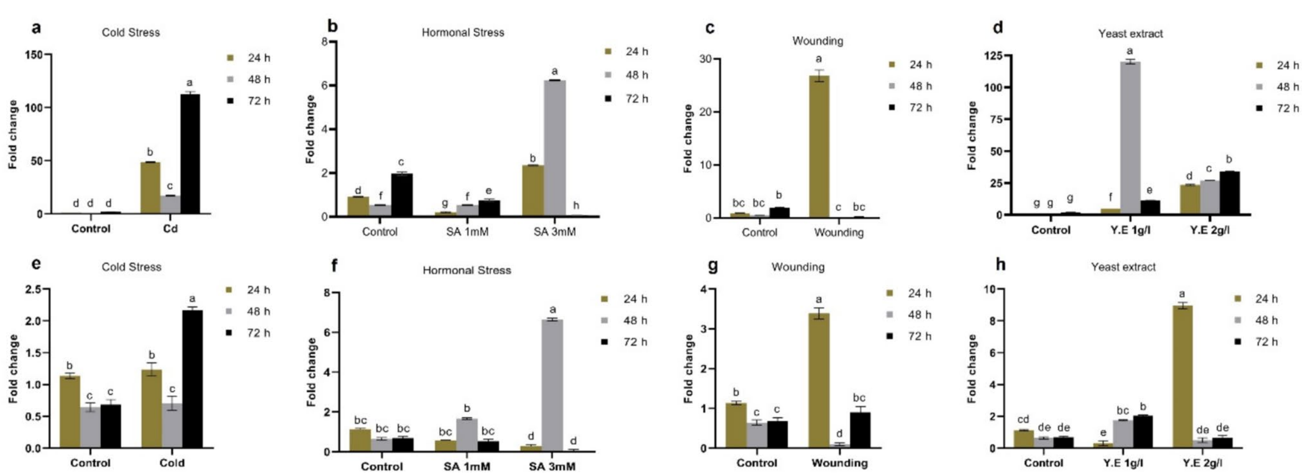

Fig. 8 Expression analysis of *CrTIDS* genes in response to stress treatments: **a–d** *CrGPPS-01* and **e–h** *CrGPPS-04*. Different letters in a bar graph show significant differences among the mean values

Previous genome-wide identification studies have reported varying numbers of *TIDS* gene homologs across different plant species. For instance, 17 *TIDS* genes were identified in *Arabidopsis* (Beck et al. 2013), 25 in *Gossypium hirsutum* (Ali et al. 2020), 14 in *Gossypium raimondi*i (Feng et al. 2023), and 10 in both *Cinnamomum camphora* (Yang et al. 2021) and *Brassica oleracea* (Yan et al. 2023). In contrast, only 5 *TIDS* genes have been identified in *Chimonanthus praecox* (Kamran et al. 2020). Notably, there remains a gap in detailed information regarding the *TIDS* gene family in *Catharanthus roseus*. With the availability of the *C. roseus* genome, it becomes possible to identify the paralogs of a particular gene or protein family. This study aimed to investigate the *TIDS* gene family in *C. roseus*, analyzing their expression patterns and potential functions throughout the plant's life cycle. Results showed that *C. roseus* genome contains *8 TIDS* genes. The corresponding *Cr*TIDS proteins are classified into four groups based on similarity with *At*TIDS, *Gohir*TIDS, *Soly*TIDS and *Bol*TIDS *proteins*. Previous studies have also divided the TIDS family in 4 groups in *Brassica oleracea* L. (Yan et al. 2023), *Chimonanthus praecox* (Kamran et al. 2020), and cotton (Ali et al. 2020), while 5 groups in *Cinnamomum camphora* (Yang et al. 2021; Jia et al. 2016). Among the four groups, group IV is the largest group, containing 16 TIDS proteins, including 2 *Cr*GGPPS (*Cr*GGPPS-06 and *Cr*GGPPS-08) and *Cr*GPPS-05 that represent *Cr*GPPS.LSU. Group I contains homomeric GPPS (CrGPPS-01) while group II contains 4 *Cr*TIDS proteins, including *Cr*GPPS.SSU (*Cr*GPPS-01). Furthermore, motif analysis also identified a conserved CXXXC motif from *Cr*GPPS.SSU (CrGPPS-01) and *Cr*GPPS.LSU (*Cr*GPPS-05), required for physical interaction between both subunits of the heteromeric GPPS protein. The CXXXC motif was also found in other members of group II, such as *Cr*GPPS-02 contains two copies of this, while *Cr*GGPPS-03 contains only one copy. All *Cr*TIDS proteins contain either one or multiple copies of conserved motifs FARM ($DDX_{2–4}D$) or SARM (DDXXD). These are aspartate-rich motifs required for the binding of substrate and also determine the catalytic activity of the TIDS enzymes. The present results are in accordance with a study carried out by Rai et al. (2013), who reported genes encoding heteromeric and homomeric GPPS from *C. roseus*.

Subcellular localization prediction revealed that most of the *Cr*TIDS proteins were localized in the chloroplast, while some of their members were also found in mitochondria and cytoplasm. GPPS are more commonly found in chloroplasts, while GGPPS are found in several organelles (Okada et al. 2000). Plastid localized GPPS and GGPPS proteins have been isolated from various plant species like tobacco (Orlova et al. 2009), red pepper (Wang et al. 2018), wintersweet (Kamran et al. 2020), rice (Zhou et al. 2017), tomato (Gutensohn et al. 2013), and *C. roseus* (Rai et al. 2013) while studies have also supported mitochondrial and cytosolic localization of TIDS proteins (Beck et al. 2013). The conservation of protein motifs within the same group or clade is noteworthy, as it provides a valuable reference for functional analysis. The conserved domain analysis reveals that all proteins are predicted to contain a conserved prenyl synthase domain PF00348, highlighting a common evolutionary foundation. The evolution of this multigene family contributes to the diversification of gene structures, leading to the multifunctional roles of certain genes, which is particularly important for adapting to environmental changes (Shang et al. 2013). Furthermore, an analysis of gene structure shows that the number of introns in *CrTIDS* genes ranges from 0 to 14, mirroring the variation seen in Brassica, which also exhibits a similar intron range (Yan et al. 2023). This information enhances our understanding of genetic evolution and adaptation in these species. In comparison, the number of introns varies from 0 to 12 in *C. camphora* (Yang et al. 2021) and 0 to 11 in *Chimonanthus praecox* (Kamran et al. 2020). However, most of the *CrTIDS* genes contain only 1 exon. The gene duplication events also play a key part in evolution and expansion of various gene families and also result in the production of new genes with novel functions (Moore and Purugganan 2003). The collinearity analysis of *TIDS* genes in *C. roseus* disclosed that there was one pair of tandem duplications and two pairs of segmental duplications. In the present study, all gene pairs showed ka/ks values less than 1, with the *CrGPPS-05/CrGPPS-06* gene pair having a maximum value of 0.39518437.

Gene expression pattern provides essential cues for potential gene function. To investigate the expression of *TIDS* genes in different tissues of *C. roseus*, the expression profiles of *CrTIDS* genes were analyzed using available transcriptomic data. The results suggested that most *CrTIDS* genes showed expression in all targeted tissues, however, *CrGGPPS-03* and *CrGPPS-02* showed tissue-specific expression. *CrGGPPS-03* showed higher expression in the stem, while *CrGPPS-02* showed higher expression in the flower, root, and leaf samples. In aerial green parts, GGPPS catalyzes the synthesis of primary metabolites. In *Nelumbo nucifera*, overexpression of *NnGGPPS1* improves carotenoid and chlorophyll content and also enhances plant growth (Dong et al. 2023a, b). *Dunaliella* species with higher GGPPS activity also accumulate high beta-carotene content (Xu et al. 2022). The *TIDS* gene family in *Arabidopsis* also showed tissue-specific expression, with *At2 g23800* (*GGPPS4*) showing maximum expression in flowers, while the transcripts of At3 g14530 (GGPPS6) were abundant in roots and siliques. However, *GGPPS11* (*At4 g36810*) was present in high abundance in all organs and tissues and is mainly responsible for most metabolites in *A. thaliana* (Beck et al. 2013). Similarly, the wintersweet *CpGPPS.SSU1* and *CpGGPPS1* showed elevated transcript levels in floral parts (Kamran et al. 2020). The expression analysis of

C. camphora reflected that *CcTIDS1* and *CcTIDS2* genes were specifically expressed in root and stem, while other *CcTIDS genes* (*CcTIDS3*, *CcTIDS9*, *CcTIDS8*, and *CcTIDS10*) showed improved expression levels in roots (Yang et al. 2021). The expression of *Panax ginseng GGPPS* (*PgGGPPS1* and *PgGGPPS2*) was higher in aerial parts and was induced by biotic and abiotic stresses (Rahimi et al. 2015).

The expression of *CrTIDS* was also observed in cell suspension and hairy root cultures treated with methyl jasmonate and yeast extract. Yeast extract was found to be more effective in inducing the expression of *CrTIDS* genes. These results were further confirmed by qRT-PCR data. The expression of *CrGPPS* appears to be induced by SA treatment also, particularly with high concentrations of SA and greater exposure time. Salicylic acid induces the expression of key genes of terpenoid indole alkaloid pathways (Soltani et al. 2020). The limiting role of *CrGPPS* was also identified by Rai et al. (2013), and they found that the expression level of *CrGPPS* in leaves is lower than in flowers and stem cells. Transient overexpression of *CrGPPS* in leaves results in 45% higher production of vindoline content. Furthermore, application of MeJ on *C. roseus* seedlings also induces expression of *CrGPPS* transcripts. The promoter analysis of the *TIDS* gene family in *C. roseus* reveals a complex regulatory network involving multiple cis-acting elements associated with various stress responses. The diversity of these elements across different *CrTIDS* genes suggests that they may be differentially regulated under distinct environmental conditions. The presence of ABRE, DRE, G-box, MYB/MYC binding sites, and other stress-related elements supports the hypothesis that *CrTIDS* genes play a significant role in the plant's adaptive responses to abiotic stresses, potentially influencing the biosynthesis of TIAs.

While this study offers a comprehensive genome-wide analysis and expression profiling of the *TIDS* gene family in *Catharanthus roseus*, the lack of functional validation through overexpression or VIGS limits our understanding of their direct roles in MIA biosynthesis. Future investigations should aim to functionally characterize key *TIDS* genes involved in the biosynthesis of pharmacologically important metabolites and correlate their expression with metabolite accumulation. Despite these limitations, the findings from this study offer a valuable genomic and transcriptomic resource, laying the groundwork for future research aimed at metabolic engineering and biotechnological enhancement of MIA production in *C. roseus*.

Conclusion

The expression profiling under stress conditions validated the differential response of *CrGPPS-01* and *CrGPPS-04*. A stress-inducible expression of *CrGPPS-01* and *CrGPPS-04* was observed in the current study; nonetheless, the expression of *CrGPPS-01* was much more pronounced compared to *CrGPPS-04*. Based on this, it was concluded that members of the *CrTIDS* gene family may play a significant role in stress responses. While the study does not provide a direct conclusion on the role of *CrTIDS* genes in the biosynthesis of biologically active phytochemicals such as MIAs, functional characterization of the identified genes can reveal their precise role in stress conditions and biosynthesis of MIAs.

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Data Availability Data is provided within the manuscript or supplementary information files.

Declarations

Conflict of interest The authors declare no competing interests.

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